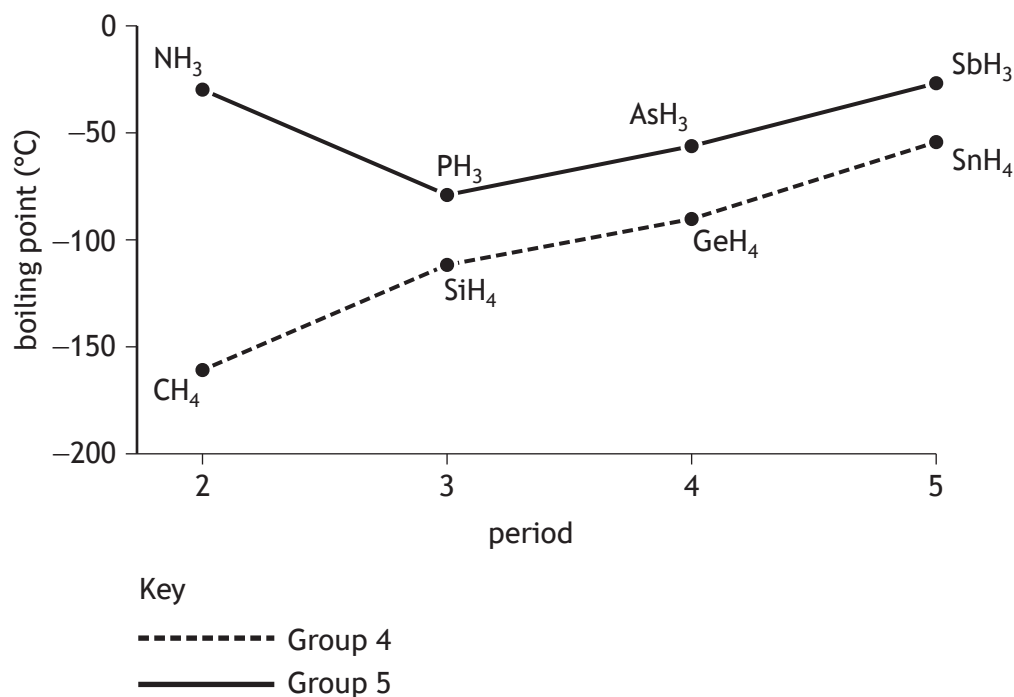


2. Hydrides are compounds containing hydrogen and one other element.

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(a) The graph shows the boiling points of group 4 and group 5 hydrides.



(i) Explain **fully** why the boiling points of the group 4 hydrides increase going down the group.

In your answer you should refer to the intermolecular forces involved.

3

(ii) Name the type of intermolecular force that is responsible for the anomalous boiling point of ammonia, NH₃.

1

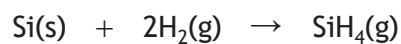
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2. (continued)

- (b) (i) Silicon hydride, SiH_4 , can be formed by reacting silicon with hydrogen.



Calculate the enthalpy change, in kJ mol^{-1} , for this reaction using the following information.

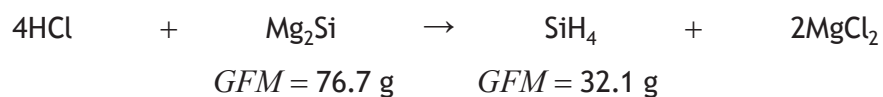
2



2. (b) (continued)

- (ii) One method of preparing silicon hydride involves reacting magnesium silicide, Mg_2Si , with hydrochloric acid, HCl .

15.32 g of magnesium silicide was reacted with excess hydrochloric acid.
2.56 g of silicon hydride was produced.



Calculate the percentage yield of silicon hydride.

2

- (iii) The table shows the melting points of silicon hydride, SiH_4 , and silicon oxide, SiO_2 .

	Melting point ($^{\circ}\text{C}$)
SiH_4	-185
SiO_2	1710

Explain **fully** why silicon oxide has a much higher melting point than silicon hydride.

3

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